

PROJECTIONS OF PLANE SURFACES

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EDGE RESTING – [GENERAL] PROBLEMS

[1] An equilateral triangular lamina of 25mm sides lies with one of its edges on HP such that

1st Position T V

the surface of the lamina is inclined to HP at 60° . The edge on which it rests is inclined to VP

2nd Position F V

3rd Position T V

at 50° . Draw its projections.

[2] A square lamina of 40mm side rests on one of its sides on HP. The lamina makes 30° to

1st Position T V

2nd Position F V

HP and the side on which it rests makes 45° to VP. Draw its projections. [Sketch book problem]

3rd Position T V

[3] A pentagonal lamina of edges 25mm is resting on HP with one of its sides such that the

1st Position T V

surface makes an angle of 60° with HP. The edge on which it rests is inclined at 45° to VP.

2nd Position F V

3rd Position T V

Draw its projections.

[4] A hexagonal lamina of sides 30mm rests on one of its sides on HP. The lamina makes 45°

1st Position T V

2nd Position F V

to HP and the side on which it rests makes 30° to VP. Draw its projections. [Sketch book problem]

3rd Position T V

[5] A rectangular lamina of sides $20\text{mm} \times 30\text{mm}$ rests on HP on one of its longer edges. The

1st Position T V

lamina is tilted about the edge on which it rests till its plane surface is inclined to HP at 45° .

2nd Position F V

The edge on which it rests is inclined at 30° to VP. Draw its projections of the lamina.

3rd Position T V

EDGE RESTING – [SPECIAL] PROBLEMS

[6] A pentagonal lamina of sides 25mm is resting on one of its edges on HP with corner

1st Position T V

opposite to that edge touching VP. The edge parallel to VP and the corner, which touches

3rd Position T V

VP is at a height of 15mm above HP. Draw the projections of the lamina. [Sketch book problem]

2nd Position F V

[7] A pentagonal lamina of sides 25mm is having a side both on HP and VP. The corner

1st Position T V

3rd Position T V

opposite to the side on which it rest is 15 mm above HP. Draw the top and front views of the

2nd Position F V

lamina.

[8] A regular hexagonal lamina of side 30mm is lying in such a way that one of its side touches both the reference planes. If the side opposite to the side on which it rests is 45mm above HP. Draw the projections of the lamina.

1st Position T V

3rd Position T V

2nd Position F V

[9] A regular hexagonal lamina of 30mm sides lying in such a way that one of its sides touches both the reference planes. If the lamina makes 60^0 with HP. Draw the projections of the lamina.

1st Position T V

3rd Position T V

2nd Position F V

[10] A regular hexagonal lamina of 30mm sides lying in such a way that one of its sides on

1st Position T V

HP while the side opposite to the side on which it rests on VP. If the lamina makes 60° to

3rd Position T V

2nd Position F V

HP. Draw the projections of the lamina.

[11] A rectangular lamina of 35mm × 20mm rests on HP on one of its shorter edges. The

1st Position T V

lamina is rotated about the edge on which it rests till it appears as a square in the top view.

2nd Position T V

The edge on which the lamina rests being parallel to both HP and VP. Draw its projections

3rd Position T V

and find its inclinations to HP and VP. [Sketch book problem]

CORNER RESTING – [GENERAL] PROBLEMS

[1] A triangular plane lamina of sides 25mm is resting on HP with one of its corners touching

1st Position T V

it, such that opposite to the corner on which it rests is 15mm above HP and makes an angle

2nd Position F V

of 30^0 with VP. Draw the top and front views in this position. Also determine the inclination

3rd Position T V

of the reference plane. [Sketch book problem]

[2] A pentagonal lamina of edges 25mm is resting on HP with one of its corners such that
1st Position T V

the surface makes an angle of 60^0 with HP. The two of the edges containing the corner on
2nd Position F V

which the lamina rests makes equal inclinations with HP. When the edge opposite to the
corner makes an angle of 45^0 with VP and nearer to the observer. Draw the top and front
3rd Position T V

views of the lamina in this position.

[3] A regular hexagonal lamina of sides 25mm is lying in such a way that one of its corners
1st Position T V

on HP while the corner opposite to the corner on which it rests on VP. If the lamina makes
3rd Position T V

60^0 to HP. Draw the projections of the lamina.

2nd Position F V

[BETA] PROBLEMS

[1] A pentagonal lamina having edges 25mm is placed on one of its corners on HP such that
1st Position T V

the perpendicular bisector of the edge passing through the corners on which the lamina rests
is inclined at 30° to HP and 45° to VP. Draw the projections of the lamina. [Sketch book problem]

2nd Position F V

3rd Position T V

[2] A regular hexagonal lamina of sides 25mm is lying in such a way that one of its corners
1st Position T V

on HP. The lamina makes 45° to HP and the diagonal passing through the corner on which
2nd Position F V

it rests is inclined at 30° to VP. Draw the projections of the lamina.
3rd Position T V

[3] A square plate of 30mm sides rests on HP such that one of its diagonals is inclined at 30°

1st Position T V

2nd Position F V

to HP and 45° to VP. Draw its projections. [Sketch book problem]

3rd Position T V

[4] A square lamina ABCD of 30mm side rests on corner C such that the diagonal AC

1st Position T V

appears to be at 50° to VP. The two sides BC and CD containing the corners C make equal

3rd Position T V

inclination with HP. The surface of the lamina makes 30° with HP. Draw its projections.

2nd Position F V

[5] A pentagonal lamina having edges 25mm is placed on one of its corners on HP such that

1st Position T V

the surface makes an angle of 30° with HP and perpendicular bisector of the edge passing

2nd Position F V

through the corner on which the lamina rests appears to be inclined at 30° to VP. Draw its

3rd Position T V

projections.

[6] A hexagonal lamina of sides 25mm rests on one of its corners on HP. The lamina makes

1st Position T V

45° to HP and the diagonal passing the corner on which it rests appears to be inclined at 30°

2nd Position F V

3rd Position T V

to VP. Draw its projections. . [Sketch book problem]

CIRCULAR LAMINA - PROBLEMS

[1] A circular lamina of 50mm diameter is standing with one of its points on the rim on HP

1st Position T V

and the lamina is inclined at 45° to HP. The diameter at right angle to the diameter which is

2nd Position F V

passing through the point on which the lamina rests is parallel to VP. Draw its projections.

3rd Position T V

[Sketch book problem]

[2] Draw the projections of a circular plate of negligible thickness 50mm diameter resting

1st Position T V

on HP on a point A on the circumference, with its plane inclined at 45° to HP and the top

2nd Position F V

view of the diameter passing through the resting point makes 60° with VP.

3rd Position T V

[3] A circular lamina of 50mm diameter rests on HP such that one of its diameter passing

1st Position T V

through the resting point is inclined at 30° to VP and 45° to HP. Draw its projections.

3rd Position T V

2nd Position F V

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